D03 · Maintenance & Asset Reliability

02 · Predictive maintenance & IoT sensors

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1 WHAT IT IS

Predictive maintenance (PdM) is the discipline of using condition-monitoring data — vibration, temperature, oil analysis, ultrasonic, motor current signature — to predict equipment failure before it occurs, enabling intervention in a planned window rather than during a forced outage. The IoT sensor layer is the data-acquisition tier; the analytics layer (rules-based thresholds, statistical anomaly detection, deep-learning models) is where prediction happens; the workflow layer (CMMS integration, planned work-order generation, technician dispatch) is where prediction becomes prevention. All three layers must exist; missing any one converts PdM from prevention into expensive observation.

2 WHY IT MATTERS

Predictive maintenance is the most over-marketed and most under-implemented maintenance discipline of the last decade — and the gap between IoT sensor deployment counts and actual reliability gains has become the dominant pattern in industrial operations. **The dominant failure mode is sensors deployed without a criticality strategy, without analytics capability, and without operational workflow — producing terabytes of data that nobody monitors, alerts that nobody triages, and dashboards that nobody consults. The failures the data could have predicted continue to occur**. The foundational text on predictive maintenance is explicit about the three-layer requirement **(Mobley, 2002 — peer-reviewed, "An Introduction to Predictive Maintenance", 2nd ed., Butterworth-Heinemann)**: PdM delivers measurable reliability gains **only** when sensor strategy is driven by equipment criticality, analytics has the capability to convert raw data into actionable predictions, and operational workflow ensures predictions trigger planned interventions. Industry data confirms — **well-implemented PdM programs reduce unplanned downtime 25–45% and extend equipment life 20–40%; poorly-implemented programs (sensors-only, no analytics or workflow) show no measurable gain despite 6-figure investments**. The asymmetry is structural, and it lives in which layers were built versus which were skipped.

3 HOW IT IS DONE

1. **Run a criticality analysis before sensoring anything — most operations have 5–15% of equipment that warrants PdM and 85–95% that does not.** Criticality is a function of (a) consequence of failure (downtime cost, safety impact, quality impact, environmental impact) and (b) probability of failure under current maintenance strategy. PdM is economically justified on assets where consequence × probability exceeds the PdM investment threshold; on lower-criticality assets, simpler strategies (run-to-failure with adequate spares, time-based PM) win. Plants that sensor everything indiscriminately produce data exhaust; plants that sensor the critical 10% with depth produce reliability gains.
2. **Choose the sensing modality to the failure mode — not to the vendor offering.** Vibration sensors for rotating equipment with bearing/imbalance failure modes; thermal imaging for electrical and friction failure modes; oil analysis for hydraulic, gearbox, and lubrication system failures; ultrasonic for compressed air leaks and electrical partial discharge; motor current signature analysis for motor and load-coupled failures. Each modality detects a specific failure-mode signature; deploying vibration sensors on equipment whose failures are thermal, or thermal cameras on equipment whose failures are bearing-driven, produces data that does not predict the relevant failures.
3. **Build or buy the analytics capability before scaling the sensor deployment — a sensor without analytics is just an expensive dial gauge.** Rules-based thresholds (vibration above X, temperature above Y) catch obvious degradation but miss complex multi-variable failure signatures; statistical anomaly detection catches drift from baseline; deep-learning models trained on labeled failure data catch specific failure-mode signatures with weeks of lead time. The analytics maturity must match the failure-mode complexity. Plants that deploy 200 sensors against a 3-rule threshold engine produce 199 false alarms and 1 missed failure per quarter.
4. **Integrate PdM alerts into the CMMS/EAM workflow — predictions that do not trigger planned work orders are predictions wasted.** The PdM platform must integrate with the CMMS such that a high-confidence prediction automatically generates a planned work order assigned to the appropriate technician with the diagnosed failure mode, required parts, and recommended intervention window. Without this integration, alerts pile up in a separate PdM dashboard that gets reviewed

monthly while the predicted failures continue to occur as unplanned events.

5. **Train reliability engineers to interpret PdM data — analytics produces a probability, not a decision.** A vibration-analytics system flagging a pump as 73% likely to fail within 4 weeks does not tell the reliability engineer whether to replace the pump, replace just the bearing, schedule a rebuild during the next planned outage, or continue monitoring. The interpretation requires reliability-engineering judgment combined with operational context (criticality, available redundancy, planned outage timing, spare parts availability). Plants that treat the analytics output as the decision produce poor decisions; plants that treat it as the input to a reliability-engineering decision produce sound ones.
6. **Measure the PdM program by failures prevented — not by sensors deployed.** The vanity metrics (sensor count, data volume, alerts generated, dashboard views) measure activity; the economic metric (unplanned failures prevented vs same period prior year, on-time-to-prediction interventions, planned-to-unplanned ratio change) measures the value the program is creating. Plants that report sensor count to leadership produce sensor expansion; plants that report failures prevented produce reliability improvement.

WORKED EXAMPLE — ILLUSTRATIVE

A petrochemical plant deploys **47 wireless vibration and temperature sensors on critical pumps and compressors** as part of a $680K PdM initiative. After 18 months: **3 critical equipment failures occurred that the deployed sensor data, reviewed in retrospect, showed clear precursor signatures for — but no one had been actively monitoring the data, no alerting workflow had been built, and the PdM "dashboard" was opened approximately once per quarter**. The total downtime from the 3 failures was 28 days at $340K/day. The remediation: **(1) criticality screening reduces the active PdM scope from 47 assets to 14 truly critical assets where the economic case justifies the program; (2) deployment of an AI-driven analytics platform (Augury) trained on industry-standard failure-mode signatures for centrifugal pumps and reciprocating compressors; (3) integration with the plant CMMS such that high-confidence predictions auto-generate planned work orders; (4) a dedicated reliability engineer with PdM-platform certification owns daily review of alerts and triage**. Results 14 months after remediation: **0 unplanned failures on the 14 monitored critical assets (vs 3 in the prior 18 months on the broader set), MTBF on monitored critical equipment +130%, total downtime cost avoided estimated at $4.8M, 12 high-confidence predictions converted into planned interventions during scheduled outages**. The sensor count dropped (47 -> 14); the reliability gain emerged because the analytics and workflow layers were built where they had been missing.

IN PRACTICE

Run criticality analysis before sensoring anything

Sensors deployed against non-critical assets generate data exhaust nobody monitors; sensors deployed against critical assets without first ranking failure-mode consequences attack symptoms rather than the highest-impact failures. Criticality analysis determines which 5–15% of assets warrant the analytics investment.

Match sensing modality to failure mode, not to vendor offering

Vibration for bearings, oil analysis for lubrication wear, motor-current signature for electrical-mechanical coupling, thermography for heat-pattern anomalies, ultrasonic for early-stage leak detection **(Mobley, 2002 — peer-reviewed, Butterworth-Heinemann)**. The vendor that sells one modality will recommend that modality for every failure.

Integrate alerts into the CMMS so predictions trigger planned work orders

A prediction that does not generate a work order is a prediction that did not happen. The workflow layer — CMMS integration, automated work-order generation, technician dispatch, predicted-failure date driving scheduling — is what converts prediction into prevention. Without it the alert lives in a dashboard nobody monitors.

Reliability engineering owns all three layers, not just sensors

Maintenance engineering owns sensor deployment; IT/OT owns platform infrastructure; nobody owns analytics or workflow integration. The structural fix is a reliability-engineering function with explicit cross-layer ownership, senior enough to negotiate across maintenance, operations, IT, and engineering.

WHAT YOU WOULD USE

PREDICTIVE MAINTENANCE STACK

Augury / Senseye (Siemens) / SparkCognition / Aspen Mtell — AI-driven PdM platforms with pre-trained failure-mode models for common rotating equipment (pumps, motors, compressors, gearboxes); the right fit for operations that need analytics capability without building data science in-house **(Augury — vendor; Siemens — vendor; Aspen Technology — vendor)**.

Emerson AMS / GE Bently Nevada System 1 / SKF @ptitude — traditional condition monitoring — Enterprise condition-monitoring platforms with deep installed bases in process industries — strong for vibration analysis on rotating equipment, with integration into broader Emerson/GE/SKF instrumentation ecosystems **(Emerson — vendor; GE Vernova — vendor; SKF — vendor)**.

AWS IoT SiteWise / Azure IoT Hub / PTC ThingWorx — IoT platform infrastructure — Cloud-based IoT platforms for sensor data ingestion, edge processing, time-series storage, and integration with analytics and CMMS; the platform layer when the PdM stack is built in-house rather than purchased turnkey **(AWS — vendor; Microsoft — vendor; PTC — vendor)**.

Targeted handheld vibration analyzer + reliability-engineer-led monthly route + CMMS integration — For operations where continuous monitoring is not justified — a portable analyzer (Emerson CSI 2140, SKF CMU 100) used on a monthly route by a trained reliability engineer, with results captured in CMMS and trended over time. The high-discipline manual option produces solid reliability gains on operations where the 24/7 monitoring economics do not justify continuous PdM.

4 SO WHAT

IoT sensors deployed without a criticality strategy, analytics capability, and operational workflow are expensive surveillance that does not prevent failures — the data gets generated, nobody uses it, the failures keep happening. Run criticality analysis before sensoring anything, match sensing modality to failure mode not to vendor offering, build or buy analytics capability proportional to failure-mode complexity, integrate alerts into the CMMS workflow so predictions trigger planned work orders, train reliability engineers to interpret outputs as inputs to engineering judgment, and measure the program by failures prevented not by sensors deployed. The three layers (sensor, analytics, workflow) are mutually dependent — and the difference between PdM programs that reduce unplanned downtime 25–45% and PdM programs that produce no measurable gain lives entirely in which layers were built versus skipped.

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